

Competitive Analysis of Hyperscale Compute Capacity Using the ‘GW-GPU-month’ Model

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TLDR

This report introduces the ‘compute-month’ (GW-GPU-Month): a functional unit for converting data center inputs - power, time, and GPU type - into the associated AI outputs in terms of training, inference, and market value. It uses cumulative GW-H100-months (‘distance’) through 2029 to forecast, price, and compare the competitive positions of frontier AI labs.

Key insights:

- **1 GW for 1 month** (730 GWh) powering Nvidia H100’s at maximum density produces ~477M H100-hours ($\sim 1.7 \times 10^{27}$ FLOPs) — enough to train **3 GPT-4** class models or **0.2 GPT-4.5** class models end-to-end, or serve ~840M frontier-model inference queries.
- Using spot market rates from **Ornn’s** H100 index, in **Jan 2026** this compute capacity was worth **\$843M**
- Compute comes from **owned capacity** (dedicated facilities tracked by **EpochAI**) plus **auxiliary capacity** (rented from cloud providers, limited public information about deal terms).
- For owned capacity by the end of 2029, Google leads Meta on raw power (101 vs 90 GW-months) — but after **adjusting for GPU** generation, Meta leads 137 to 117 GW-H100-months. **OpenAI+Microsoft** reaches **299 GW-H100-months** — a **162 GW-H100-month lead** over the nearest competitor (Meta).
- At its planned **10 GW** capacity, **OpenAI’s Stargate** could replicate the entire cumulative compute from **GPT-3 through GPT-5** (~ 24.6 GW-Months over **2020–2025**) in under **2.5 months** — illustrating how high-speed facilities compress years of historical compute distance into weeks
- Strategic partnerships like **Crusoe’s \$1.25B** purchase of turbines from **Boom Supersonic** capable of generating **1.2GW** captures **~24 compute-months** that competitors cannot access due to supply chain constraints. Using H100 baseline, ROI is between $\sim \$20B$ (spot value) and **~\$7.8B** (with depreciation).
- These gaps are not apples-to-apples. OpenAI and Google build for **mass-market inference at 900M+ MAU**; Meta and xAI embed AI into **ecosystems of billions of users**; Anthropic operates at 1/7th the owned capacity of its nearest competitor, betting on post-training and enterprise trust over raw compute

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1) The AI Race: Speed, Distance, and Acceleration

The AI race has a simple premise: whoever secures the most compute capacity the soonest wins. What "winning" means — AGI, market dominance, GDP growth — depends on who you ask, but the imperative to build does not. The scaling laws imply that model quality is proportional to the compute capacity used for training, making the data center buildout a literal race. And like any race, position is determined by distance, speed, and acceleration.

Speed is a lab's maximum operational capacity at a given point in time — how many gigawatts of power are running GPUs right now. The speedometer.

Distance is cumulative capacity utilized over time — the integral of the speed curve. The odometer. Distance determines how many models have been trained, how many responses provided to users, how many experiments have been run, how much frontier AI learnings have compounded.

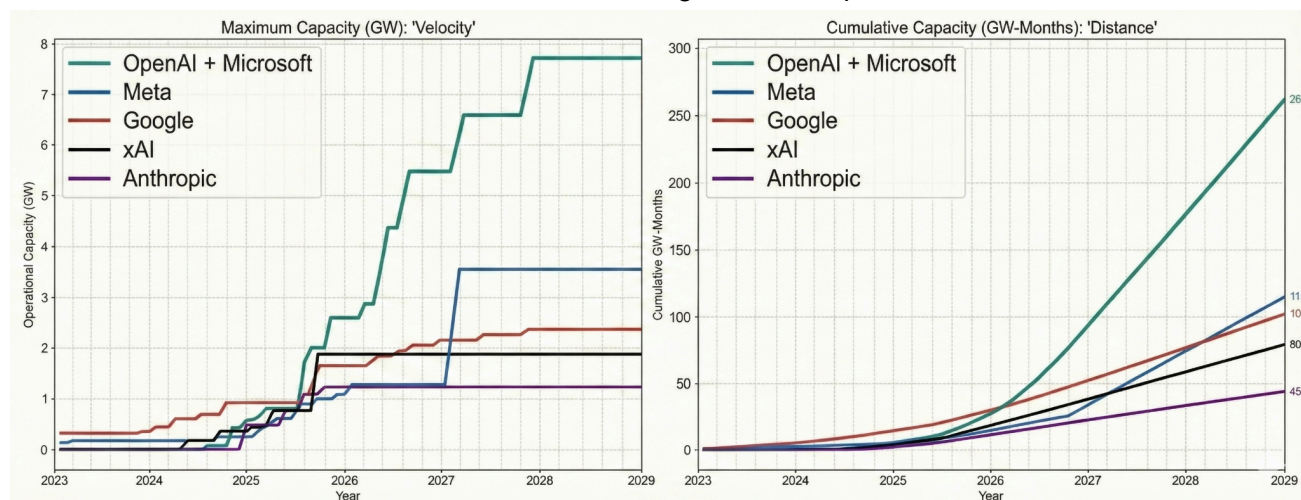
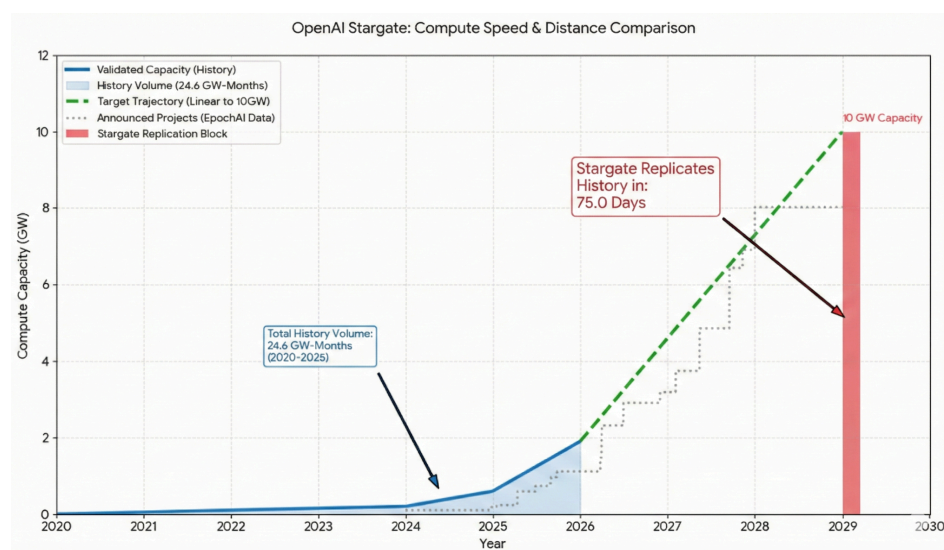


Figure 1: Left: "Maximum Capacity" showing GW over time for frontier labs (2023–2030). Right: "Cumulative Capacity" showing GW-Months accumulated over the same period. Data from EpochAI, aggregation and charts by author.

While Meta is projected to surpass Google's *speed* in Q127, it takes an additional year to overtake Google's *distance*. The difference between speed and distance is starkest at scale. Consider OpenAI's Stargate project:



At its planned **10 GW** capacity, OpenAI Stargate could replicate the entire training frontier from **GPT-3 to GPT-5 (2020–2025)** — approximately 24.6 GW-Months of cumulative compute — in **under 2.5 months**. Accounting for the improved performance of newer GPU generations, that window shrinks further by a factor of >2.

Figure 2: OpenAI Stargate compute velocity: equating historical and future distance over time

Acceleration - the rate at which new capacity comes online - varies significantly across labs. Per analysis from EpochAI, hyperscale data centers are taking between 1 and 3.6 years from construction start to achieving 1 GW of total facility power. In the wake of this unprecedented speed & scale development, new projects are expected to take between 3 and 7 years from pre-development to first billable token. This time variance across roadmaps is critical for evaluating competitive positioning. *"Time to power is not a nuisance. It is the asset."* — Suhail Y Tayeb, *Dirt to Data* newsletter (Jan 2026)

2) GW-GPU-month Model

Comparing development paths across labs requires factoring in different power sources, timelines, and GPU models. *Which path delivers more competitive advantage in terms of raw compute capacity over time?*

Answering that requires a functional unit: a concept from the Department of Energy's lifecycle assessment methodology that measures a system based on what it produces. Compute is a flow unit, analogous to seat-miles in aviation or room-nights at a hotel - capacity that isn't used when available is lost permanently.

A standard compute-month (GW-H100-Month) represents the output of the maximum number of Nvidia H100-equivalent GPUs powered by 1 GW for one month, at 2025 data center efficiency standards.

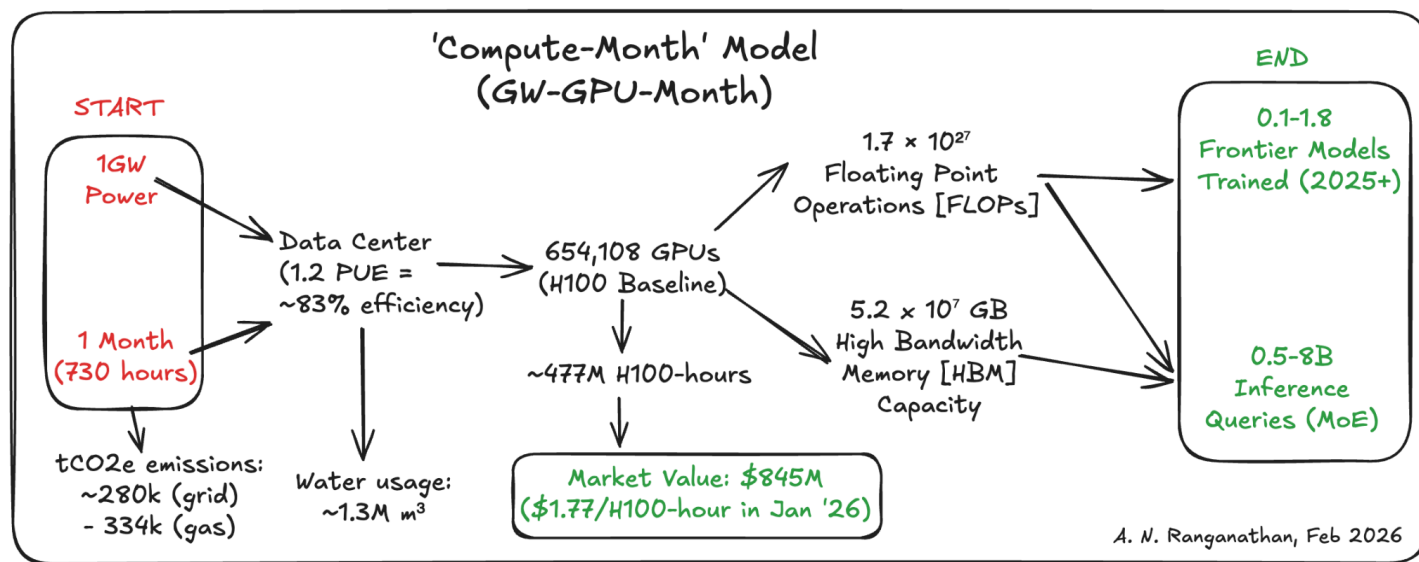


Figure 3: "Compute-Month" Framework flow diagram. Full derivation included in the appendix.

This model scales to other GPUs using scaling factors based on thermal density (# GPUs per GW), peak FLOPs (# models trained per month), and memory bandwidth (# inference queries per month). For example, the Nvidia B200 has **2.53x** higher peak FLOPs and **2.39x** higher memory bandwidth, but 1.71x higher power draw than the H100, resulting in **x0.58** GPUs per GW (381,563).

- For Training: 1 GW-B200-Month = **1.47** GW-H100-Month
- For Inference: 1 GW-B200-Month = **1.39** GW-H100-Month

Why the H100 baseline?: It catalyzed the current buildout (GPT-4, Claude 3) and wave of AI investment, represented ~49% of Nvidia GPU shipments in 2023–2024, and has the longest spot market price history.

Technical Benchmarks: What One Compute-Month Produces

Models Trained: Using training compute estimates from EpochAI dataset (<https://epoch.ai/data/ai-models>):

Model	Release Date	Est. Training FLOPs	Est. GW-H100-Months to Train	Equivalents per 1 Compute-Month
OpenAI GPT-4	3/15/23	2.10E+25	0.34	2.9
xAI Grok-2	8/13/24	2.96E+25	0.48	2.1
Claude Sonnet 3.7	2/24/25	3.00E+25	0.55	1.8
OpenAI GPT-4.5	2/27/25	3.80E+26	6.21	0.2
Meta Llama Behemoth	4/5/25	5.18E+25	0.85	1.2
xAI Grok 4	7/9/25	5.00E+26	8.17	0.1
DeepSeek-V3.2-Exp	9/29/25	4.18E+24	0.07	14.6

Raw training FLOPs from EpochAI; R&D overhead from EpochAI [report](#) on OpenAI 2024 compute allocation; assumes facility utilization of 90% and Model Flop Utilization (MFU) of 40%, detailed in Appendix.

Inference Served: throughput varies dramatically by model size, architecture, and workload:

Workload Tier	Model Profile (# parameters & architecture)	Workload Profile (Tokens in/out)	Queries per GW-H100-Month
Heavy	Frontier models, ~1.2T parameters, MoE active e.g. Apple-Google deal (Bloomberg)	W10k: 11,500 tokens/query	~0.15 billion
Medium	Older frontier models, ~500B parameters, MoE active	W1k: 2,000 tokens/query	~2 billion
Light	Open-source models, <50B parameters, dense	W1k: 2,000 tokens/query	~20 billion

Parameter estimates from EpochAI; Inference workloads from Artificial Analysis; methodology in Appendix

As the industry shifts from training-heavy to inference-dominated workloads driven by agents, real-time applications, and enterprise deployment, the inference benchmarks become increasingly important for valuing capacity. (See “[Memory: How It Works and Why It Sets the Cost of Compute](#)” - Ornn)

3) Competitive Analysis

The AI compute market operates in two tiers. 1) Owned capacity dedicated hyperscale facilities that labs build and control; and 2) auxiliary capacity, rented from cloud providers to fill the gap between what a lab owns and what its roadmap demands. The interplay between these tiers defines the competitive landscape: owned capacity sets the floor, auxiliary capacity sets the ceiling.

Owned Capacity

Figure 1 showed the simplistic GW-month view — however this does not factor in the GPU model. Using data from EpochAI timelines for density and H100-equivalencies provides a more accurate perspective.

Note: forecasts update as facilities break ground. For example, xAI Mississippi (Jan 9, 2026) and OpenAI-Oracle New Mexico (Jan 26, 2026) are not yet reflected in EpochAI's satellite-verified dataset.

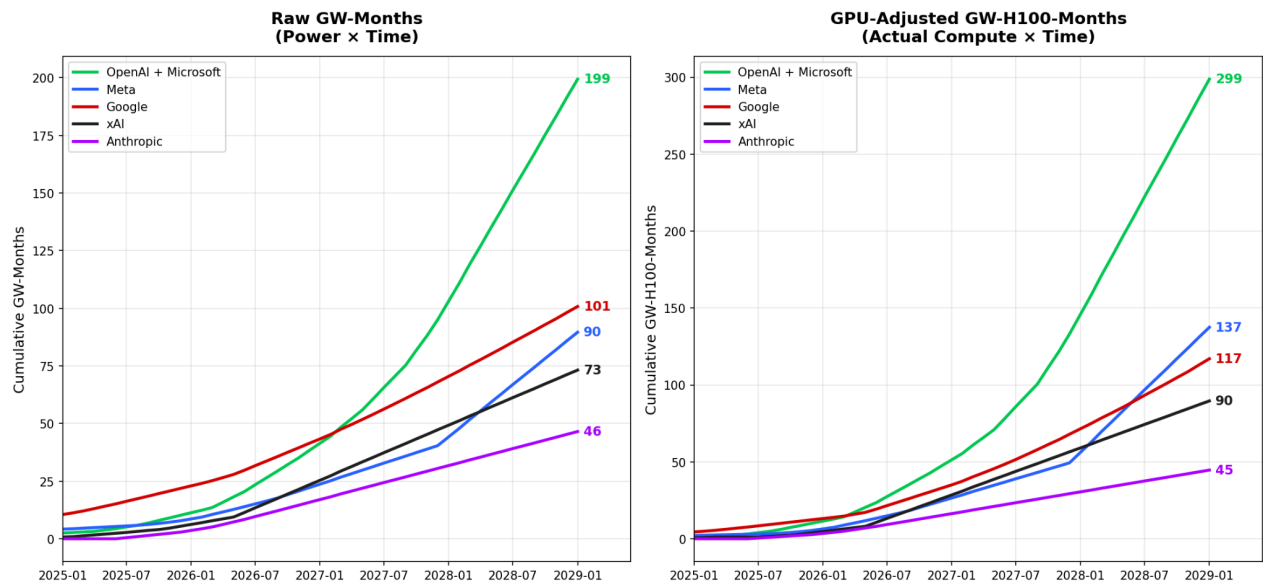


Figure 4: Left — Raw cumulative GW-Months (power × time only). Right — GPU-adjusted cumulative GW-H100-Months incorporating hardware density and performance multipliers. Data from EpochAI, aggregation and charts by author.

Meta and Google illustrate why GPU adjustment matters. On raw power & time, Google actually leads Meta (101 vs. 90 GW-months) — but after adjusting for chip generation, Meta leads 137 to 117, a 17% gap that represents a reversal in ranking driven entirely by next-gen hardware. OpenAI+Microsoft's lead widens similarly, from 2.0× to 2.6× over Google once chip generations are factored in. Anthropic primarily rents auxiliary compute from cloud providers, so the GPU-adjusted cumulative compute from its single owned facility remains roughly flat.

The competitive picture, however, is not just about who has the most compute. Each lab's target market shapes how much capacity they need, what kind of workloads dominate, and where on the speed/distance curve they need to be.

Criteria	OpenAI + MSFT	Google	Meta	xAI	Anthropic	DeepSeek
Cum. GW-months	288	95	131	130	60	Not tracked by EpochAI
Cum. GW-H100-months	439	163	212	118	58	Est. <5
Density uplift (vs H100 baseline)	1.53×	~0.6× (TPU fleet avg.)	1.62× (Hyperion next-gen)	~0.9×	~1.0×	N/A (constrained hardware)
Monthly Active Users	~900M	~750M (+Apple Siri)	~3.9B (ecosystem)	<100M	~30M (enterprise)	~100M (est. mid-2025)
Primary Workload	Inference at scale	Inference + training	Recommendation / content generation	Content generation / inference	API / enterprise inference	Training efficiency R&D; inference

This comparison of owned capacity highlights the strategic bet each lab is making:

- OpenAI and Google are building for mass-market uncapped inference at 900M+ MAU scale, where cumulative distance functions as a serving-capacity moat
- Meta is the biggest beneficiary of GPU adjustment (90→137) despite its slower timeline. Labs building for existing ecosystems can afford to wait for better GPUs; labs racing for frontier performance cannot.
- Anthropic has bet that post-training optimization and enterprise trust can substitute for infrastructure scale, operating at 1/7th the owned capacity of its nearest competitor.

The GPU adjustment sharpens these distinctions: it widens OpenAI's lead, reveals Meta's hidden hardware advantage over Google, and confirms that Anthropic's position is a strategic choice rather than a deficit.

4) Valuing the Gaps

The speed/distance framing indicates who's ahead - but not what that lead is worth, what it would cost to close, or how the rental market connects labs with distance deficits to providers with auxiliary capacity. Answering those questions requires converting GW-GPU-months into priced output.

Auxiliary Capacity

EpochAI tracks dedicated hyperscale facilities — call these owned capacity, analogous to proven reserves in oil & gas. But labs also rent auxiliary capacity from cloud providers at negotiated rates to supplement what their owned infrastructure can't supply, just as oil majors purchase crude on the spot market when reserves can't meet throughput. Several major deals illustrate how labs rent auxiliary compute to close speed gaps.

- Google Cloud rents to Meta (>\$10B, undisclosed timeline, Aug 2025)
- Google Cloud rents to OpenAI (undisclosed details, May 2025)
- Google Cloud rents to Anthropic (>1GW, 1M TPUs, undisclosed timeline, updated Oct 2025)
- CoreWeave rents to OpenAI (>\$22.4B over 5 years, updated Sep 2025)
- Microsoft Azure rents to Anthropic (~1GW, undisclosed timeline, Nov 2025)
- Oracle rents to xAI (\$10B, undisclosed details, May 2024)

Financial terms are rarely disclosed in full, which makes it difficult to accurately incorporate each labs' auxiliary compute into the 'distance' forecast. However, these partnerships illustrate the total addressable market for auxiliary compute providers: every compute-month a lab doesn't own is a potential purchase order for AWS, GCP, CoreWeave, Nebius, or other neoclouds.

Ornn Index & GPU Depreciation

Since 1 GW-H100-month = ~477.5 million H100-hours, the market value can be derived using price data from Ornn. At Jan 2026 spot rates (\$1.77/H100-hour), one GW-H100-month ≈ \$845 million.

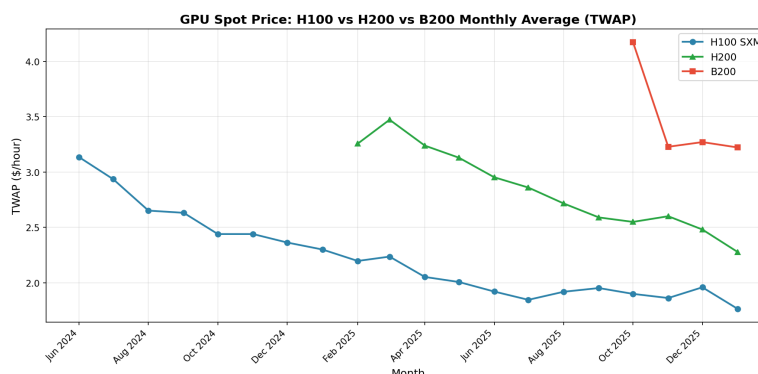


Figure 5: H100 vs H200 vs B200 Monthly Avg. Prices. Data from Ornn API; stats & plot prepared by author

H100 price history from June 2024 through early 2026 implies ~27% annual depreciation, consistent with a market in “backwardation”: a H100-month generated today is worth substantially more than one generated in 2028 because the same hardware commands a higher market rate earlier in its lifecycle.

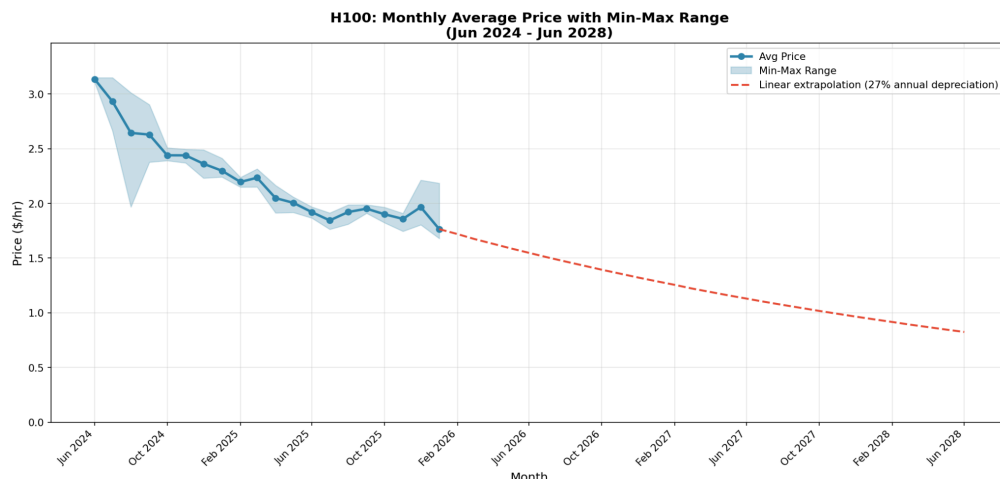


Figure 6: Historical price range of H100 and best-fit linear extrapolation of 27% annual depreciation. Data from Ornn API; stats & plot prepared by author

This depreciation curve also prices delay. A 1 GW facility delayed by 12 months at current rates forgoes ~\$9.5B in present compute value that can never be recovered — and the market will have moved to lower price points by the time operations begin. This asymmetry makes construction timeline risk the single largest financial variable in data center investment. A full analysis of project delays and operational interventions that mitigate them (cooling innovation, load flexibility, organizational friction) is forthcoming as a separate report.

Cost of Catching Up

The distance deficit framing reduces competitive catch-up to a rates problem. Using the competitive analysis of owned capacity: if xAI trails Meta by 94 GW-H100-Months by the start of 2030, then additional capacity must come from auxiliary providers or accelerated construction. For perspective, the largest known auxiliary deal is CoreWeave-OpenAI at ~\$22.4B over five years, estimated to deliver roughly 2–3 GW-H100-Months per month.

To close the gap in 48 months, xAI needs ~2 additional GW-H100-Months per month of sustained output.

On H100 infrastructure:

- Sustained additional capacity: ~2.0 GW (~1.3M GPUs)
- Monthly spot value: $2.0 \times \$843\text{M} = \text{\textbf{\$1.69B/month}}$
- Extrapolated total through 2030 (no depreciation): ~\$81B

On B200 infrastructure (1,200W liquid-cooled):

- Sustained additional capacity: ~1.3 GW (~503K GPUs), producing the same H100-equivalent output at 1.47× efficiency
- 1 GW-B200-Month = 278.6M B200-hours; at Jan '26 spot rate of \$3.22/hr (Ornn), spot value = \$898M
- Monthly cost for 1.3 GW: $1.3 \times \$898\text{M} = \text{\textbf{\$1.17B/month}}$
- Extrapolated total through 2030 (no depreciation): ~\$56B

The B200 path is ~30% cheaper despite a higher hourly rate (\$3.22 vs. \$1.77) because each B200 GPU produces more H100-equivalent compute per dollar. The effective cost per GW-H100-Month of output is ~\$611M on B200 versus \$843M on H100 — however given the limited price history for B200, this relationship is subject to rapid change.

Shortening the timeline makes the problem disproportionately more expensive. Closing the same 94 GW-H100-Month gap in 24 months instead of 48 requires double the sustained capacity, but the total cost is more than double: with H100 rates depreciating ~27% annually, compute generated in 2026 commands ~50% higher spot rates than compute generated in 2028. Urgency carries a premium.

Valuing Strategic Partnerships

Renting auxiliary compute capacity in perpetuity is not feasible for closing large gaps. Strategic partnerships that accelerate owned capacity timelines represent a structurally different lever than renting auxiliary compute: they shift the speed curve itself rather than purchasing distance at spot rates.

Case Study: Crusoe × Boom Supersonic: In December 2025, Crusoe announced a \$1.25B order of gas turbines from Boom Supersonic — 29 turbines at 42 MW each, generating over 1.2 GW of behind-the-meter power. Gas turbine procurement from GE Vernova, Siemens, or Mitsubishi currently requires 36–48 months, with deep backlogs. By securing alternative supply through Boom's aerospace-grade manufacturing, Crusoe locked in capacity that competitors cannot replicate during this procurement window.

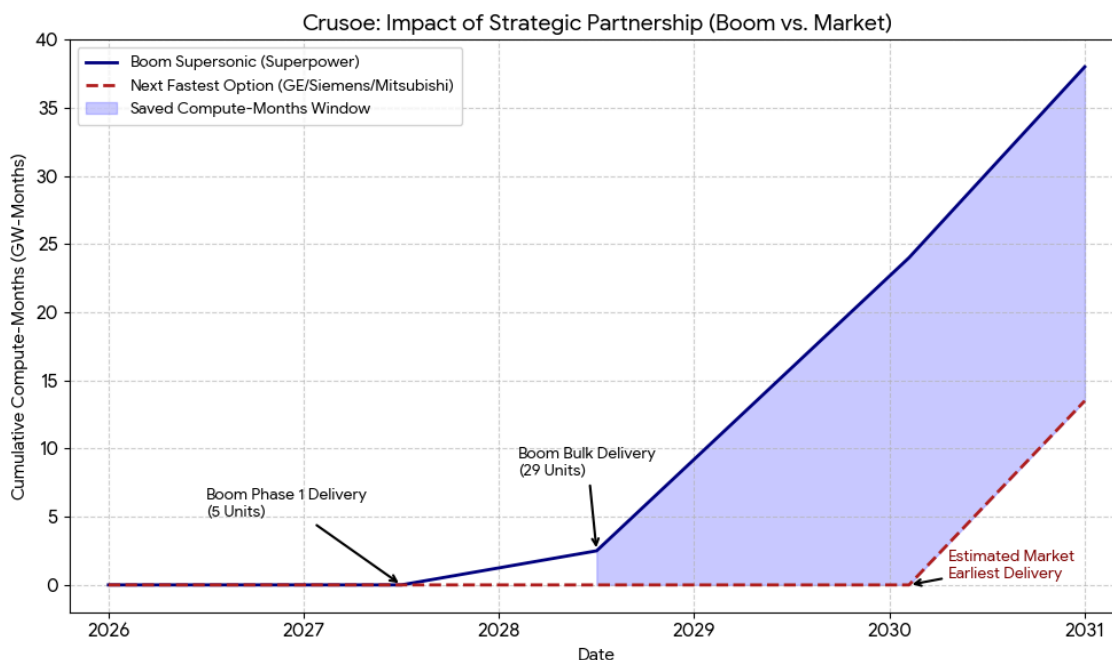


Figure 7: GW-month plot of incremental capacity gained by Crusoe through turbine deal with Boom; prepared by author

The \$1.25B translates to ~24 accelerated compute-months:

- ~\$20B at spot rates — 16× ROI
- ~\$7.78B cumulative revenue at 27% annual depreciation — 6.2× ROI
- 18.5 GPT-5 training runs or 24 Claude Sonnet 4 equivalents

Quantifying the ROI of strategic partnerships that accelerate timelines is arguably the most novel use case of the compute-month model.

5) Conclusion

Speed and distance both matter in the AI infrastructure race — but they matter differently. Speed determines what a lab can do today; distance determines what it has learned, trained, and shipped over time. The labs pulling ahead are not simply those with the most capacity at any given moment, but those whose capacity came online earliest (acceleration) and has been compounding the longest.

The GW-H100-Month model offers a functional unit to quantify this. Energy developers can evaluate projects in terms of AI outputs rather than megawatts alone. AI practitioners can translate infrastructure timelines into training and inference capacity across GPU generations. Policymakers assessing permitting timelines or grid interconnection queues can price what delay costs in forgone compute value. The unit bridges these stakeholders by converting between shared physical inputs and the outputs each cares about.

Pricing that output requires market data, and Ornn's GPU spot index provides the only continuous public price signal across GPU generations. As the rental market matures and new hardware comes online, maintaining this analysis will depend on sustained price transparency — making indices like Ornn increasingly important infrastructure in their own right.

The race is early and the track is long - but every lap counts.

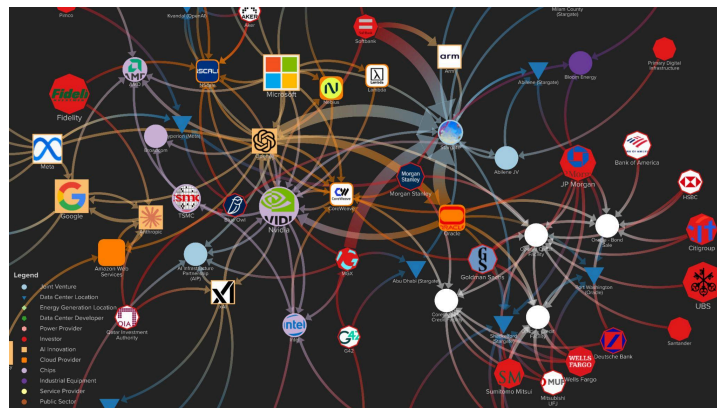
Future work:

- Delay Mitigation: how cooling innovation (e.g. from Crucible Capital 'Building a Datacenter: Part 2'), load flexibility (e.g. EmeraldAI), organizational friction (research from Prof. Alex Newkirk) and engagement strategies (insights from Suhail Y Tayeb) affect timelines
- Inference weighted compute-months (GW-HBM-month): measuring capacity by serving throughput rather than training FLOPs, as the industry shifts toward agentic inference-dominant workloads and Ornn's DRAM price indexes develop a longer price history

About the Author

Aditya Nirvaan Ranganathan is an MBA candidate at UChicago Booth (class of 2027). His work focuses on the intersection of energy infrastructure and AI compute economics. Previously a software engineer in capital markets at Jefferies, he held leadership roles at startups spanning clean energy (Flowcarbon, Elective Power) and blockchain infrastructure (Celo). He holds a B.A. in Music and Physics from Swarthmore College.

Other work



[AI Infrastructure Ecosystem Map](#): network analysis of 200+ entities and 300+ relationships across AI labs, hardware manufacturers, energy providers, investors, other stakeholders

Sources

- EpochAI — Frontier model training compute [dataset](#); data center timeline [dataset](#) (satellite-verified)
 - Ornn — H100 GPU spot price index (monthly TWAP)
 - Crucible Capital — Building a Datacenter Part 1 [report](#)
 - Nvidia — H100 SXM reference architecture [specifications](#)
 - Suhail Y Tayeb — “What actually breaks first in AI data center projects?” [article](#) (Dirt to Data, Jan 2026)
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Appendix

Assumptions & Derivations of GW-H100-month

Assumptions:

- Power Usage Effectiveness (PUE): 1.2 (industry average for hyperscale facilities, meaning ~83% of total power reaches IT equipment)
- Utilization factor of 90%: accounts for hardware failures, job scheduling gaps, maintenance windows, and checkpointing downtime
- IT overhead multiplier: 1.82× (accounts for networking, storage, etc.)
- Model Flop Utilization (MFU): 40% (GPT-4 was reported at 34%; assume modest improvement)
- R&D overhead: 10x based on EpochAI [report](#) about OpenAI 2024 compute allocation
- Inference efficiency factor: $\eta = 0.70$ (accounts for KV-cache, kernel overhead, batching inefficiency)

Derivation for Training FLOPs:

- $1 \text{ GW} / 1.2 \text{ PUE} / 1.82 \text{ IT overhead} / 0.7 \text{ kW per H100} = 654,108 \text{ H100 GPUs}$
- $654,108 \text{ H100} \times 730 \text{ hours} = 477,498,840 \text{ H100-hours}$
- H100 peak FLOPs for BF16 dense training = 989.5×10^{12}
- $989.5 \times 10^{12} \text{ FLOP/s} \times 3,600 \text{ s/hr} = 3.5622 \times 10^{18} \text{ FLOP per H100-hour}$
- $477,498,840 \text{ H100-hours} \times 3.5622 \times 10^{18} \text{ FLOP/H100-hour} = 1.70 \times 10^{27} \text{ FLOPs}$
- $1.70 \times 10^{27} \text{ FLOP} \times 90\% \text{ utilization} \times 30\% \text{ MFU} \times 0.1 \text{ R\&D overhead} = 6.12 \times 10^{25} \text{ effective FLOPs}$

Methodology for Inference queries:

- Active weight size: $W = N_{\text{active}} \times 2 \text{ bytes} \rightarrow \text{convert to TB}$
- Tokens/sec/GPU: $r = (3.35 \text{ TB/s} \div W) \times 0.70 (\eta)$
- Tokens/month (cluster): $r \times 654,108 \text{ GPUs} \times 2,628,000 \text{ sec/month}$
- Queries/month: $\text{tokens/month} \div \text{tokens/query}$ (using W1k or W10k)

Limitations

- Does not consider rack density as a constraint
- Does not consider non-fungibility of compute including region, uptime guarantees, etc.